AeroTrack

Air quality monitoring with Apache Hop

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Module 6 Final Project | Apache Hop

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You cannot see the air you breathe.

Air pollution is one of the world's largest environmental health risks. CO and NO2 levels can spike for hours without anyone noticing, until the exposure has already happened.

The real question is not “is the air polluted”. It is: exactly WHEN was it worst, and what conditions made it worse?

Without treated data, that question has no answer

Raw sensor noise

Missing readings, a -200 sentinel value and empty export rows hide the real signal.

No structure

Nothing tells you which hours, days or months were actually critical.

No memory

Without a database, every analysis starts from zero, every single time.

AeroTrack was built to close exactly this gap.

AeroTrack

Air quality monitoring with Apache Hop

From raw sensor data to a clear, trustworthy answer.

Built end to end with Apache Hop

THE DATA

Dataset and problem statement

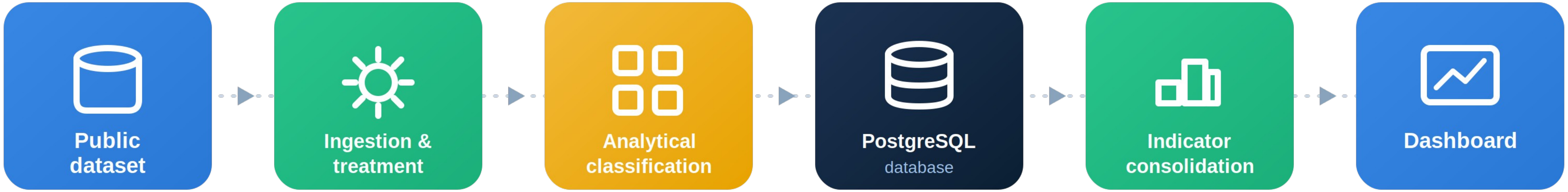
- Dataset: Air Quality, UCI Machine Learning Repository
- Source: a monitoring station in an Italian city (De Vito et al.)
- Period: March 2004 to February 2005
- Volume: 9357 hourly readings, 15 variables

PROBLEM

In which periods were the worst air quality conditions recorded, and which environmental variables were associated with them?

THE SOLUTION

End-to-end architecture



Two Apache Hop pipelines, orchestrated by a workflow: pl_01_ingestion_treatment and pl_02_indicator_consolidation, feeding a PostgreSQL database and a live dashboard.

Ingestion and treatment pipeline

- CSV Input reads the raw dataset (semicolon separator, comma decimal)
- Filter Rows removes the empty rows left by the dataset export
- Formula builds a single date-time text; Null If handles the -200 missing-value sentinel
- Select Values renames fields and converts text into a proper Date
- Calculator, Value Mapper and Number Range derive year, month, time of day and CO/NO2 level
- Janino flags whether a reading is valid; Filter Rows splits valid from rejected readings
- Insert/Update loads PostgreSQL keyed on data_hora, safe to rerun

Consolidation pipeline and orchestration

pl_02_indicator_consolidation

- Table Input aggregates treated readings by day and by month
- Insert/Update writes resumo_diario_qualidade_ar and resumo_mensal_qualidade_ar

wf_air_quality_orchestration

```
START → Ingestion and treatment → Indicator consolidation → Process completed
```

The second hop only runs if ingestion succeeds, preventing indicators from ever being built on an incomplete load.

Treatments applied

- -200 sentinel: converted to null before any calculation (Null If)
- 114 empty export rows: dropped during ingestion (Filter Rows)
- Readings with no reference pollutant (CO, NOx and NO2): separated into leituras_rejeitadas
- CO and NO2 analytical classification (GOOD, MODERATE, CRITICAL): calibrated from the dataset's own percentiles, with no external regulatory reference

PERSISTENCE

Database and rerun safety

`leituras_qualidade_ar`	Treated, detailed data, one row per valid reading
`leituras_rejeitadas`	Readings with no reference pollutant
`resumo_diario_qualidade_ar`	Indicators aggregated by day
`resumo_mensal_qualidade_ar`	Indicators aggregated by month

Rerun without duplicates

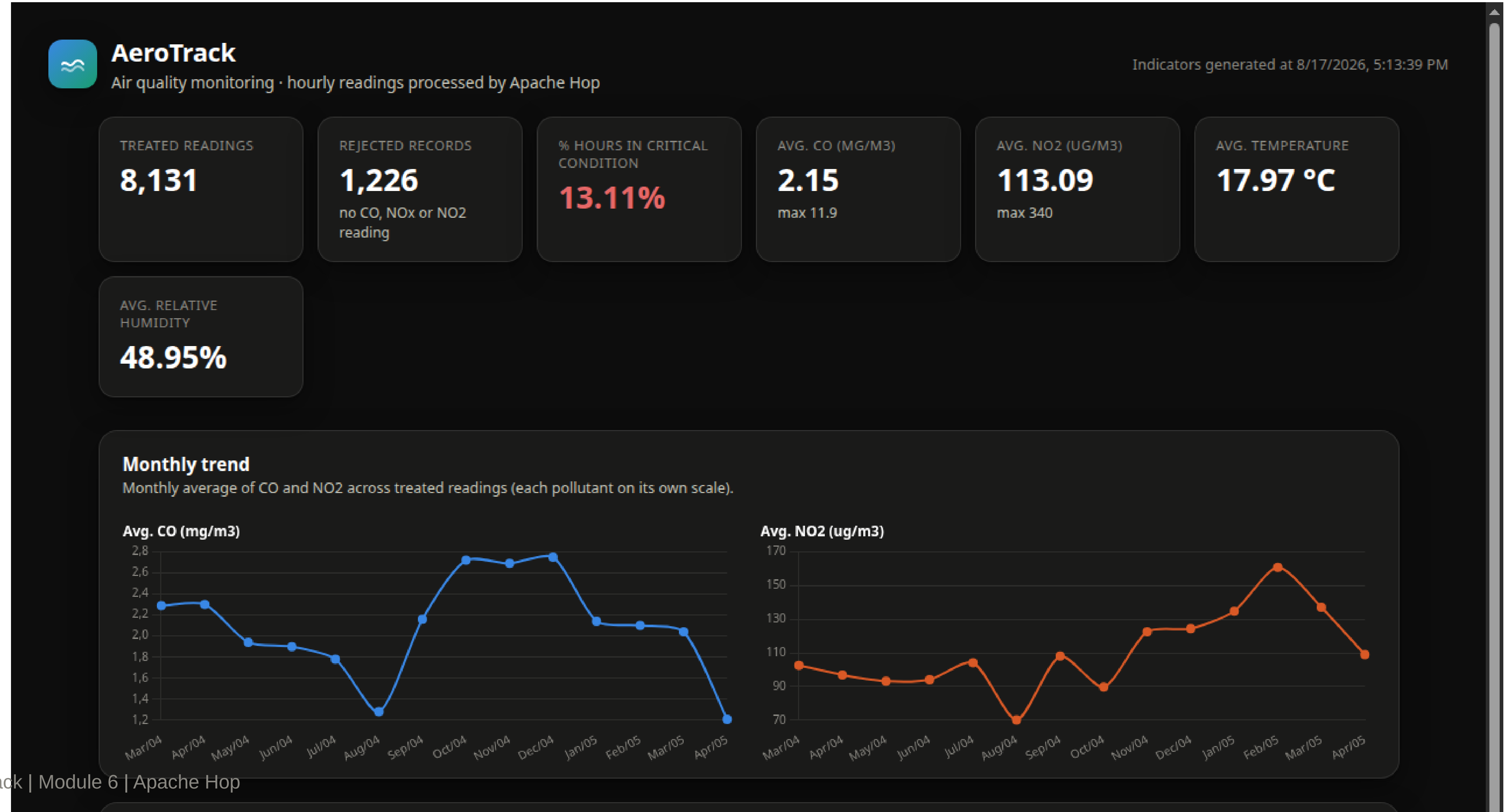
UNIQUE key on data_hora plus Insert/Update: rerunning the process updates existing records instead of duplicating. Validated by running the workflow twice in a row (8131 rows both times).

Indicators calculated

- Total treated and rejected readings
- Percentage of hours in critical condition
- Average and maximum CO and NO2
- Monthly trend of CO and NO2
- Distribution of readings by level (good, moderate, critical)
- Readings by time of day
- Ranking of the ten worst days

VISUALIZATION

AeroTrack dashboard



ANSWERING THE QUESTION

Ranking of the ten worst days

Ranking of the 10 worst days

Days sorted by the highest daily average CO, with average NO2 and percentage of hours in critical condition.

DATE	AVG. CO	AVG. NO2	CRITICAL HOURS	% CRITICAL HOURS
10/19/2004	5.65	139.75	6	75%
12/23/2004	5.32	166.52	17	70.83%
11/22/2004	5.06	163	15	62.5%
11/25/2004	4.64	159.74	13	54.17%
12/15/2004	4.43	162.96	13	54.17%
11/1/2004	4.33	102.22	11	45.83%
4/14/2004	4.19	130.53	5	33.33%
10/6/2004	4.19	98.38	4	50%
2/10/2005	4.16	223.35	16	66.67%
12/22/2004	4.06	172.09	10	41.67%

Validation of results

TREATED READINGS	REJECTED RECORDS	% CRITICAL HOURS
8,131	1,226	13.11%

AVG. CO	AVG. NO2	DAYS CONSOLIDATED
2.15 mg/m3	113.09 ug/m3	363

Valid plus rejected readings match the dataset total after removing the 114 empty export rows. The worst-day ranking shows high NO2 and a high critical-hour share on the same dates as the highest CO, consistent with real pollutant behavior.

Every critical hour now has a record

- AeroTrack turns raw, noisy sensor exports into a database that can answer, with evidence, when air quality was worst
- The CO/NO2 classification was calibrated on the dataset's real distribution, not arbitrary thresholds
- Every table is safe to reload: Insert/Update guarantees no duplicate history, run after run

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Thank you